

LLM: Inside of Model

model.h

```
ModuleList(
  (0-11): 12 x GPT2Block(
    (ln_1): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
    (attn): GPT2Attention(
      (c_attn): Conv1D(nf=2304, nx=768)
      (c_proj): Conv1D(nf=768, nx=768)
      (attn_dropout): Dropout(p=0.1, inplace=False)
      (resid_dropout): Dropout(p=0.1, inplace=False)
    )
    (ln_2): LayerNorm((768,), eps=1e-05, elementwise_affine=True)
    (mlp): GPT2MLP(
      (c_fc): Conv1D(nf=3072, nx=768)
      (c_proj): Conv1D(nf=768, nx=3072)
      (act): NewGELUActivation()
      (dropout): Dropout(p=0.1, inplace=False)
    )
  )
)
```

modelni **hidden-layer** i nimalardan tashkil topganini kurish uchun model.attribute deb yozamiz. Misol GPT-2 model uzini **12 ta hidden layeri** borakan

✗ istalgan **modelni hidden-layerini kurish uchun** hardoim model.h yozilmaydi. Hammani uzini **mahsus attribute bor**. Misol bazi modellarda model.layers deb istash kere buladi

elementwise_affine=True - bu degani model faqat normalization qilib raqamlarga utkazib quymasdan ularni manosi buyicha normalization qiladi

wte – word token embedding

wpe – word position embedding

wpe ni hozrgi frontier modellar ishlatishadi. u berilgan **suzning position** buyicha oladi

Contextualization: **attention** + **FFN** + **LayerNorm** + **Residual connections** – shularni barchasi bilan bulib raqamlarni **wpe dan foydalangan** holda **bir-biri bilan boglash**.

Max_token: **tokenlar sonini cheklash**. Yani, tokenlarni sonini berilgan limitgacha chiqaradi.

EOS: ketma-ketlik tugaganini bildiruvchi unique token

Qisqacha architecture:

1. "Should I learn AI?" – degan input qiymatni berdik

2. ["Should", "I", "learn", "A", "I"] – shunday holda modelimiz tokenlarga ajratadi
3. [0.242, 0.432, 0.624, "1.445", "-0.132"] – ajratilgan tokenlarni embeddings qilib vectorlarga uzgartiradi
4. Contextualization buyicha vectorlarni yani usha raqamlarni wpe yordamida bir-biri bilan boglaydi.
5. Possible outputs – eng confidence kup outputni tanlaydi. Agar EOS bulsa javobni tugatadi. Va albatta javobni max_token ga ahimiyat bergan holda javob beradi